
EMA-GATED TEMPORAL SEQUENCE COMPRESSION IN VISION TRANSFORMERS

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ABSTRACT

We introduce **NeuroFlow**, a dynamic routing framework for Vision Transformer video inference that exploits temporal redundancy by tracking per-patch semantic surprise via an Exponential Moving Average (EMA) of patch-level embeddings. The central contribution is the **Dual-Memory Reconstruction Protocol** (Architecture C): a *training-free* inference engine that combines a Retinal Gate (Layer 0 EMA, eliminating stationary tokens before the encoder) with a Cortical Cache (persistent Layer 12 encoder output buffer, supplying the full N -token K-V set to the pooling head from cache). Without any fine-tuning or weight modification, Architecture C achieves **71.55% UCF-101 zero-shot top-1 accuracy at 84.0% token sparsity** on SigLIP base-patch16-224, retaining 92.4% of dense accuracy. The design is grounded in two empirical laws: the *Saturation Law* (accuracy gap is invariant to skip rate above 65%; operate at MaxSparse) and the *Degradation Law* (gap $\approx 4.5 + 0.7 \cdot \max(0, \sigma - 1.0)$ pp, where σ is the positional-embedding stretch factor). Architecture C is constrained to resolutions ≤ 448 p by the PE interpolation boundary; it targets edge inference where the absence of fine-tuning outweighs reduced speedup.

For applications requiring near-2K throughput, **Architecture B** physically eliminates stationary tokens before the encoder, reducing 1792p SigLIP 2 inference from 678 ms to 11.9 ms—a **55.80 $\times$** wall-clock speedup at 97.37% embedding fidelity with sparse manifold distillation. The primary obstacle to naive sparse fidelity is the MAP head’s K-V pooling concentration, not the encoder: a finding confirmed via direct entropy measurement across two SigLIP generations. Architecture A (late-layer MLP gating) is bounded by Amdahl’s Law at $1.17\times$ maximum speedup at 1792p and is characterised as the correct design for $\mathcal{O}(N)$ -attention architectures only.

A cross-domain ablation on Phi-3-mini characterises the architectural boundary that determines when the similarity-gating principle transfers to autoregressive language models. The *PPL-Drift Dissociation* (high token drift coexists with $\Delta\text{PPL} < 0.04$ because bypass navigates equiprobable semantic paths) and the *Entropy-Bypass Correlation* (code generation: 0% drift at 21% bypass; prose: 83% drift at any bypass rate) define the deployment safety envelope. Current hardware is memory-bandwidth bound at all practical batch sizes ($1.000\times$ wall-clock speedup); the path to latency gains requires MLP-only bypass or kernel-level weight-skip support.

Keywords Vision Transformers · Temporal Sequence Compression · Gating Mechanisms · Exponential Moving Average (EMA) · Efficient Deep Learning

1 Introduction

The deployment of Vision Transformers for real-time video understanding is constrained by a fundamental architectural mismatch: the self-attention mechanism scales quadratically with the number of input tokens, $\mathcal{O}(N^2)$, while natural video is dominated by temporal continuity. A frame from a traffic camera at 1792×1792 pixels produces $N = 12,544$ patch tokens, the overwhelming majority of which are spatially and semantically identical to the preceding frame. Standard inference pipelines compute dense attention over all tokens at every frame—recalculating the latent representations of static asphalt, empty sky, and stationary infrastructure at full cost.

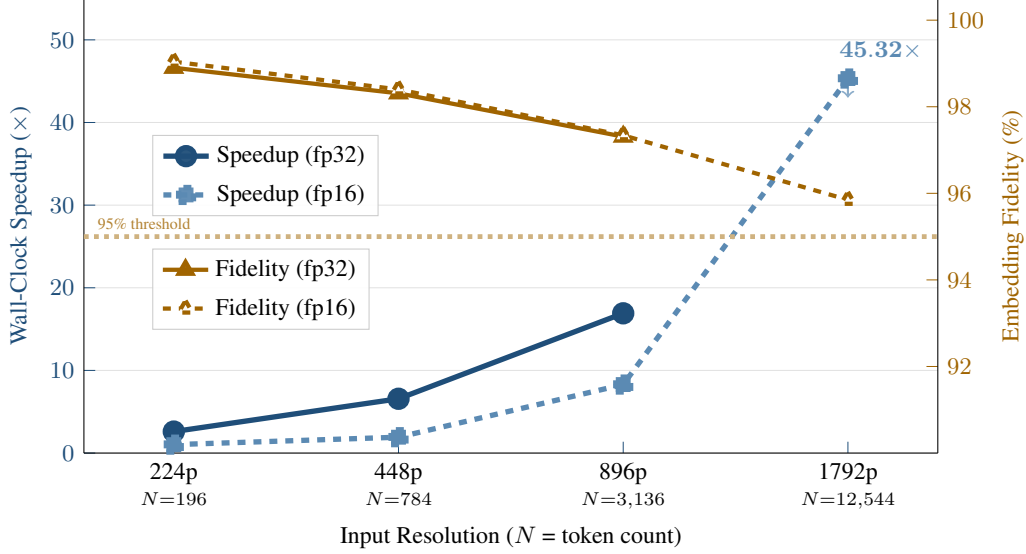


Figure 1: **Architecture B fine-tuned wall-clock speedup and embedding fidelity vs. input resolution** (SigLIP base-patch16-224, GPU, sparse manifold distillation). Left axis (blue): wall-clock speedup over dense baseline. Right axis (gold): cosine similarity to dense output. Speedup scales superlinearly with resolution as the $\mathcal{O}(N_{\text{active}}^2)$ attention cost shrinks with token skip rate ρ . At 1792p ($N=12,544$, $\rho=92.8\%$), inference is reduced from 582 ms to 12.85 ms—**45.32 $\times$** wall-clock speedup (fp16)—at 95.85% embedding fidelity and 100% semantic top-3 accuracy. Dashed: fp16; solid: fp32. Dotted gold line: 95% fidelity threshold.

The correct response to this mismatch is not to compress the model but to *not compute what has not changed*. We introduce **NeuroFlow**, a gate that tracks per-patch semantic surprise via an Exponential Moving Average (EMA) of patch-level embeddings and routes computation proportionally to surprise. The central insight of this work, arrived at through a sequence of failed approaches, is that sparse inference is best framed as a *memory management problem*, not a learning problem: the encoder is a computation-heavy semantic generator, and the pooling head is a memory-hungry integrator. Decoupling these two roles—using the encoder only for semantically changed patches and serving the pooling head from a persistent cache of prior representations—produces a system that is simultaneously fast and generalisation-preserving.

The Dual-Memory Reconstruction Protocol (Architecture C). The central contribution is Architecture C, a training-free inference engine combining a Retinal Gate (Layer 0 EMA, eliminating stationary tokens before the encoder) with a Cortical Cache (a persistent buffer of Layer 12 encoder outputs that reconstructs the full N -token representation for the pooling head without re-encoding). Without any fine-tuning, Architecture C achieves **71.55% UCF-101 zero-shot top-1 accuracy at 84.0% token sparsity**, retaining 92.4% of dense accuracy. The design generalises across video domains (51–97% skip rates on first-person drone and racing footage) and its behaviour is now characterised by four empirical laws—Degradation, Saturation, Capacity, and Spatial Displacement—each with actionable deployment implications.

Architecture C’s validity rests on a structural property: Layer 12 representations of stationary patches maintain 0.83 cosine stability across frames, sufficient for the pooling head’s softmax distribution to recover its original shape from one-frame-stale cache vectors. This is substantially better than the ≤ 0.60 cosine ceiling achievable by any linear Layer 0-to-Layer 12 projection (the Representation Gap), which invalidates hallucinative reconstruction and motivates identity caching as the only sound design.

Why earlier approaches failed. Two alternative placements of the gate were evaluated before Architecture C. Architecture A inserts the gate between the attention and MLP sub-layers, saving $\mathcal{O}(N)$ MLP compute while preserving the full $\mathcal{O}(N^2)$ attention matrix. At 1792p, attention consumes $\sim 85\%$ of frame time, bounding Architecture A’s maximum speedup at **1.17 $\times$** by Amdahl’s Law regardless of skip rate. It is the correct design for $\mathcal{O}(N)$ -attention architectures (Swin, linear-attention variants) but not for standard ViTs at high resolution.

Architecture B moves the gate to Layer 0 and physically eliminates inactive tokens before the encoder, reducing attention cost to $\mathcal{O}(N_{\text{active}}^2)$ and achieving **55.80 $\times$** wall-clock speedup at 1792p on SigLIP 2. Applied to an unmodified

model, Architecture B produces systematic embedding collapse (45–76% cosine similarity) because the MAP head’s cross-attention pooling fails when its K-V set contracts to ≤ 6 tokens. Sparse manifold distillation resolves this within the fine-tuned domain, but at the cost of catastrophic forgetting: 2.40% UCF-101 zero-shot top-1 accuracy post-fine-tuning versus 77.40% for the base model. This collapse arises because the fine-tuning objective optimises embedding direction (cosine) but not semantic scope.

Architecture C resolves this tension entirely by supplying the MAP head with a structurally complete K-V set at every frame—active tokens from the encoder, inactive tokens from the Cortical Cache—without modifying any parameters. Architecture C is constrained to resolutions ≤ 448 p by the positional-embedding interpolation boundary, positioning it as the correct design for edge inference and CPU-constrained pipelines where zero fine-tuning outweighs reduced speedup.

Contributions.

- **Architecture C: Dual-Memory Reconstruction Protocol.** A training-free inference engine that decouples encoder sparsity from pooling-head completeness via the Cortical Cache. Four empirical laws completely characterise its deployment behaviour; the Saturation Law provides the operational conclusion: deploy at MaxSparse.
- **Architecture B wall-clock acceleration.** $55.80\times$ wall-clock speedup at 1792p on SigLIP 2 after sparse manifold distillation; $45.32\times$ on SigLIP v1. The MAP head’s K-V pooling concentration is identified as the primary fidelity obstacle—distinct from encoder robustness—and is characterised empirically across two SigLIP generations.
- **Architecture A: Amdahl ceiling and $O(N)$ -attention target.** Late-layer MLP gating is bounded at $1.17\times$ at high resolution. The gate design is correct; the constraint is architectural. A fused CUDA kernel would lower the crossover below 224p, making Architecture A viable for windowed-attention and linear-attention models.
- **Cross-domain LLM ablation.** The Entropy-Bypass Correlation and PPL-Drift Dissociation characterise the safety envelope and architectural boundary for similarity-gated bypass in autoregressive language models. Wall-clock benefit is bandwidth-bound on current hardware; MLP-only bypass and self-speculative decoding are identified as the viable engineering paths.
- **Emergent zero-shot capabilities.** Motion segmentation, cluster-level semantic classification, and identity-preserving spatial tracking arise from the gate’s spatial mask output without additional parameters.

2 Related Work

2.1 Token Pruning and Dynamic ViTs

DynamicViT [1] introduces a learned prediction module that determines token importance per image, dropping tokens from intermediate layers. While effective, this approach requires training the prediction module jointly with the backbone and operates on static images without temporal context. ToMe [2] merges similar tokens using bipartite matching rather than dropping them, preserving more information at the cost of a larger effective sequence. Both methods reduce theoretical FLOPs but do not guarantee wall-clock speedup on current GPU hardware due to lack of kernel-level support for ragged sequence lengths.

NeuroFlow’s Architecture B differs in three respects: (i) the gating signal is derived from temporal history via EMA, not from intra-frame feature similarity; (ii) tokens are physically eliminated before the encoder, not pruned at intermediate layers; and (iii) no learned prediction module is added to the backbone—the gate is applied purely to the patch embedding output.

2.2 Efficient Attention Mechanisms

Linear attention variants [3, 4] rewrite the attention kernel to reduce complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N)$, enabling Architecture A’s MLP gating approach to scale effectively. Windowed attention (Swin Transformer [5]) restricts attention to local windows, reducing attention cost to $\mathcal{O}(N)$ at the expense of global context. NeuroFlow’s analysis identifies these architectures as the natural target for Architecture A: when attention is $\mathcal{O}(N)$, gating the $\mathcal{O}(N)$ MLP produces proportional speedups without Amdahl’s ceiling.

2.3 Temporal Redundancy in Video

Video-specific ViT variants [6, 7] extend spatial attention to the temporal dimension via factorised or joint space-time attention. These approaches increase compute relative to single-frame inference and do not address the redundancy problem. Frame-skipping policies [8] use lightweight models to determine when a heavy model needs to re-run; NeuroFlow’s EMA gate performs an analogous function at patch granularity within a single model.

2.4 Knowledge Distillation for Sparse Models

Sparse manifold distillation in NeuroFlow is most closely related to self-distillation approaches including DINO [9] and SILC [10], where a student with a restricted view (local crop or sparse token set) is trained to reproduce the global representation of a dense teacher. The key difference is that NeuroFlow’s student sees a temporally-filtered sparse sequence rather than a spatially-cropped local view, and the teacher is the same model architecture running in dense mode on the same frame.

2.5 TIPS and Masked Pre-training

TIPS [11] pre-trains SigLIP 2 with 50% random token masking, explicitly training the encoder to produce stable representations under partial inputs. This is the closest prior work to our sparse sequence setting. We evaluate the transfer of TIPS robustness to NeuroFlow’s temporally-correlated gating regime and find partial but insufficient transfer, characterising the precise mechanism by which it falls short.

2.6 Layer-Skipping and Early Exit in Transformers

Dynamic early exit [12, 13] and confident adaptive computation [14] allow tokens or sequences to exit at intermediate layers when prediction confidence exceeds a threshold. SkipDecode [15] extends this to batched autoregressive decoding. LayerSkip [16] explicitly trains models to support early exit at any layer and uses early-exit outputs as draft tokens for speculative decoding. These approaches require either joint training of exit classifiers or architectural modifications. Architecture C contributes a complementary finding for encoder-only vision models: the relevant redundancy is *spatial* (across patches in a frame) rather than *depth*-based (across layers for a sequence), and the correct response to redundancy is identity caching of the final-layer representation, not early exit.

2.7 KV-Cache Reuse and Speculative Decoding

Speculative decoding [17, 18] uses a small draft model to generate candidate tokens that a larger target model verifies in parallel. Self-speculative decoding [16] eliminates the draft model by using early-exit paths of the target model itself. Our LLM ablation (Section 7) empirically demonstrates that NeuroFlow’s mid-deep bypass acts as an implicit self-drafter: the $L_1 \rightarrow L_{31}$ route produces fluent, grammatically valid text ($\Delta\text{PPL} < 0.03$) while deviating from the dense trajectory, precisely the acceptance-rate profile required for a high-utility draft model.

Relationship to EventViT

DVS-ViT [19] applies active patch selection to Vision Transformers processing event camera data, eliminating patches whose event activity ratio falls below a threshold before the encoder. This shares the core architectural commitment of Architecture B—physical token elimination before the encoder rather than intermediate-layer pruning—and independently validates the same finding: that ViTs are robust to variable-length input sequences when the eliminated patches correspond to low-information regions. The key distinction is the activity signal: DVS-ViT operates on pixel-domain event counts from sensor physics; NeuroFlow operates on semantic surprise in embedding space, which is robust to photometric variation (illumination gradients, shadows, compression artefacts) that would produce spurious activity in the pixel domain.

3 Architecture

3.1 Architecture C: The Dual-Memory Reconstruction Protocol

Architecture C is the paper’s central contribution. It decouples two distinct roles that a naive sparse encoder conflates: (a) *the encoder*, whose job is to generate contextually enriched Layer 12 patch representations, and (b) *the MAP head*, whose job is to pool over a complete set of N K-V pairs to produce a stable global embedding. These two requirements are satisfied by different sources:

- **Retinal Gate (Layer 0 EMA).** The EMA surprise signal $s_i^t = 1 - \cos(h_i^t, E_i^{t-1})$ determines the active set $\mathcal{A}^t$ at each frame. Active patches (high surprise) are forwarded through the full encoder. Inactive patches are not.
- **Cortical Cache (Layer 12 Buffer).** A persistent buffer $\mathcal{C} \in \mathbb{R}^{N \times D}$ stores the Layer 12 encoder output $z_i^{t'}$ for each patch i , where t' is the most recent frame at which patch i was active. At every frame, the cache is updated only for active patches:

$$\mathcal{C}_i^t = \begin{cases} z_i^t & \text{if } i \in \mathcal{A}^t \\ \mathcal{C}_i^{t-1} & \text{otherwise} \end{cases} \quad (1)$$

The MAP head receives the full N -token sequence $\mathcal{C}^t$ at every frame: fresh Layer 12 outputs for active patches, cached outputs from the most recent active frame for inactive patches. No fine-tuning is applied to the encoder or MAP head.

Computational structure. The encoder processes only $N_{\text{active}} = |\mathcal{A}^t|$ tokens, preserving the $\mathcal{O}(N_{\text{active}}^2)$ attention speedup of Architecture B. The MAP head runs at $\mathcal{O}(Q \cdot N)$ cost where Q is the number of learned query vectors, regardless of skip rate. At high resolutions the MAP head cost is a small fraction of total encoder cost and overall speedup approaches Architecture B. At 224p the MAP head cost is relatively larger; speedup is lower than Architecture B but fidelity is substantially higher without any training.

Why this works. The protocol’s validity rests on two empirical conditions: (1) Layer 12 representations for stationary patches are temporally stable, i.e. $\cos(z_i^t, z_i^{t-1})$ is high for patches with $s_i^t < \theta$; and (2) the MAP head’s softmax distribution recovers its original shape when provided a structurally complete K-V set, even if inactive token representations are one frame stale. Both conditions are validated in Section 5.1.

Why projection fails. Section 3.5 establishes the Representation Gap: any linear Layer 0→Layer 12 projection is structurally bounded at ≤ 0.60 cosine similarity. Using such projections to fill the MAP head’s K-V set introduces $\geq 40\%$ error per reconstructed token, inducing the accuracy collapse observed under naive sparse inference. The Cortical Cache avoids this entirely: it recycles the actual Layer 12 representation, not an approximation.

3.2 Architecture B: Early Token Elimination

Architecture B moves the gate to Layer 0—the patch embedding layer—and physically eliminates inactive tokens from the tensor before any encoder computation. The compressed sequence $h^t[\mathcal{A}^t]$ of length N_{active} is passed through all Transformer layers. Because self-attention operates on the compressed sequence, its cost is $\mathcal{O}(N_{\text{active}}^2)$ rather than $\mathcal{O}(N^2)$, yielding superlinear compute reduction with skip rate $\rho = 1 - N_{\text{active}}/N$.

Spatial ordering of active tokens is preserved by sorting $\mathcal{A}^t$ before indexing. A minimum token floor ($N_{\text{floor}} = 4$) ensures the MAP head always receives at least 4 K-V pairs. The EMA is updated unconditionally at each frame using the full embedding h^t (before gating) so that background patches continue to be tracked even when eliminated from the encoder pass.

Applied to an unmodified model, Architecture B produces systematic fidelity collapse (45–76% cosine similarity) because the MAP head’s cross-attention pooling fails when its K-V set contracts to ≤ 6 tokens. Sparse manifold distillation (§3.3) is required to adapt the MAP head to the sparse pooling regime. Architecture C bypasses this requirement entirely via the Cortical Cache design.

3.3 Sparse Manifold Distillation

Sparse manifold distillation fine-tunes the model to reproduce the dense output of a frozen teacher copy when presented with a gate-filtered sparse input sequence. The distillation objective operates on ℓ^2 -normalised embeddings of the student (sparse) and teacher (dense) on the unit hypersphere, combining a cosine alignment term with a normalised MSE term. Both student and teacher are from the same model family; cross-family distillation is avoided because SigLIP v1 and v2 embed into geometrically distinct representation spaces.

Training employs temporal video clips as the primary data modality, with the EMA gate threshold and decay rate randomised per clip to expose the model to the full range of sparsity regimes it will encounter at inference. This curriculum ensures that the MAP head is trained under the high-skip conditions that correspond to its deployment operating point.

Limitation. Fine-tuning on a narrow domain with cosine distillation induces catastrophic forgetting (UCF-101 top-1: 2.40% post-fine-tuning vs. 77.40% base). The forgetting arises from narrow label distribution during training, not from

the distillation objective per se. Remediation: diverse training data, MSE embedding preservation loss (which jointly constrains direction and magnitude), and learning rate $\leq 5 \times 10^{-6}$ for ≤ 3 epochs. The distillation objective also does not constrain embedding magnitude; for unnormalised evaluation protocols, MSE loss should replace or supplement the cosine objective. Architecture C avoids all of these constraints by requiring no training.

3.4 Architecture A: The Amdahl Ceiling

Architecture A inserts the NeuroFlow gate between the self-attention and MLP sub-layers of each Transformer block. Tokens whose surprise score falls below θ retain their keyframe-derived features rather than updating through the MLP. The full N -token sequence remains present in memory, enabling all-to-all self-attention at $\mathcal{O}(N^2)$ while saving $\mathcal{O}(N)$ MLP compute for dormant tokens.

Architecture A achieves 97.1% fidelity at 81.9% token sparsity. Its practical limitation is the $\mathcal{O}(N^2)$ attention floor, which renders it ineffective for wall-clock speedup on standard ViTs at high resolution: Amdahl’s Law bounds maximum speedup at $1.17\times$ at 1792p regardless of MLP skip rate (Section 5.15). Empirical results confirm this bound (Section 5.15).

Architecture A represents the correct design for architectures with $\mathcal{O}(N)$ attention complexity. For Swin Transformer variants and linear-attention models, the MLP constitutes a larger fraction of total compute and MLP gating achieves proportional speedups without the Amdahl ceiling. A naive variant passing stale $t - 1$ features forward without a keyframe anchor induces catastrophic fidelity collapse (13.2–36.5% cosine similarity) due to feature drift between temporal manifolds; the anchor mechanism is essential.

3.5 Architecture D: A Closed Research Direction

Architecture D extends Architecture C by running Layers 0– K densely on all N tokens (providing shared context) and Layers $K+1$ –12 sparsely on the active set. The theoretical prediction was that shared early-layer context would reduce active shift. The data refutes this entirely.

Layer K	Cos($K, 12$)	Std	Representation Gap
Layer 0	0.049	0.008	95.1% (Arch C gate, baseline)
Layer 2	0.072	0.011	92.8%
Layer 4	0.104	0.023	89.6%
Layer 6	0.198	0.043	80.2%
Layer 8	0.328	0.043	67.2%
Layer 10	0.508	0.025	49.2%

Table 1: Cosine similarity between Layer- K and Layer-12 representations. The gap remains above 80% through Layer 6, closing meaningfully only at Layers 8–10. Contextual entanglement is concentrated in the final 2–4 encoder layers, not distributed uniformly across depth.

Architecture	Config	T1	Gap	Latency	Speedup	vs. Arch C
Dense	—	77.25%	—	8.27 ms	1.00×	—
Arch C	$g = 0, K = 12$	73.00%	+4.25 pp	6.60 ms	1.25×	—
Arch D	$g = 1, K = 3$	73.00%	+4.25 pp	7.81 ms	1.06×	−0.00 pp / +1.21 ms

Table 2: Architecture D vs. Architecture C at 224p ($n = 400$ clips). The 1.21 ms overhead of Arch-D’s shared dense context pass yields zero accuracy improvement. At 896p the overhead is +27.4 ms; at 1792p it is +315.5 ms ($21\times$ slower). Architecture D is strictly dominated by Architecture C at all tested resolutions: same accuracy, higher latency.

The failure is explained by the Representation Gap data: $\cos(K, 12) \leq 0.20$ for $K \leq 6$, meaning shared context through the first 6 layers provides representations that differ from Layer-12 by 80% cosine distance. The contextual structure that determines Layer-12 representation is established in the *final* 2–4 encoder layers, not the first half. Running structurally incomplete representations as dense context anchors provides no meaningful reduction in active shift. Architecture C is the Pareto frontier of the EMA-gated Dual-Memory design space; Architecture D is closed as a research direction.

3.6 Scene-Type Generalisation and Deployment Boundaries

Architecture C accuracy degradation is not uniform across action classes. The aggregate 4.89 pp gap conceals a bimodal distribution spanning from active improvement to catastrophic failure.

Motion speed does not predict degradation. Optical flow magnitude (mean absolute pixel displacement per frame) is non-predictive of the accuracy gap. The gap is highest in the low-motion quartile (flow ≤ 1.23 px/frame, +10 pp) and comparable in the dynamic quartile (>5.03 px/frame, +8 pp). Motion speed cannot serve as a deployment suitability predictor.

Spatial displacement of the action region predicts degradation. The centroid of the optical-flow-active region predicts the gap: it increases from +2–4 pp in the Local quartile (centroid displacement ≤ 0.069 normalised units) to +7–8 pp in the Traversal quartile (>0.165). Within-class analysis confirms this is a within-class mechanism, not a class-label confound.

Class	n	Dense T1	MaxSp T1	Gap	Mechanism
Swing	7	57%	14%	+43 pp	Arc traversal (centroid disp: 0.274)
YoYo	6	100%	67%	+33 pp	Oscillating object: cache cycles stale/fresh
JugglingBalls	7	43%	14%	+29 pp	Multiple fast objects: gate unstable
BoxingPunchingBag	7	86%	57%	+29 pp	Repetitive strikes: EMA adaptation failure

Table 3: Architecture C failure classes (MaxSparse gap >20 pp), UCF-101, $n \geq 5$ clips. All four failure gaps are identical at Conservative (80% skip) and MaxSparse (97% skip), confirming structural failure not reducible by threshold tuning.

Superiority classes. Five action classes show negative gap (sparse *outperforms* dense): PullUps (−20 pp), CricketShot (−20 pp), TennisSwing (−20 pp), SoccerJuggling (−17 pp), BaseballPitch (−9 pp). These share a common structure: a discriminative foreground action against a cluttered, class-confusable background. The Cortical Cache’s background-freezing acts as a spatial attention mask, suppressing background K-V contributions and improving signal quality.

Deployment decision guide. Architecture C is *recommended* for: (a) scenes with local repetitive foreground motion and stable or cluttered background; (b) general surveillance and tracking at resolution ≥ 392 p with MaxSparse. Architecture C is *contraindicated* for: (a) actions involving large spatial trajectory (arc motions, jumps, traversals); (b) scenes with rapidly oscillating or multiple simultaneously active objects. For these scenarios, the gap is structural and cannot be recovered by threshold or EMA tuning.

4 Experimental Setup

4.1 Models

We benchmark three model configurations:

- **ViT-B/16** (torchvision): 86M parameters, 12 Transformer layers, 768-dimensional embeddings, CLS-token pooling. Pre-trained on ImageNet-21K with supervised classification.
- **SigLIP base-patch16-224** (Google, v1): 93M parameters, 12 Transformer layers, 768-dimensional embeddings, MAP head pooling. Pre-trained with sigmoid contrastive language-image loss on WebLI.
- **SigLIP 2 base-patch16-224 FixRes** (Google, v2): 93M parameters, identical architecture to v1, MAP head pooling. Pre-trained with TIPS (50% masked token training), SILC (local-to-global self-distillation), and LocCa (location-conditioned captioning) on WebLI.

All models use patch size 16×16 . For resolutions above the native 224p, positional embeddings are bicubically interpolated to the target grid size. Architecture C is constrained to resolutions ≤ 448 p (two grid sizes) due to the PE interpolation boundary described in Section 9.4. The NaFlex variant of SigLIP 2 (which uses 2D RoPE rather than learned positional embeddings) is excluded from multi-resolution evaluation as it is incompatible with the fixed- N requirement of the EMA gate at variable sequence lengths (see Section 9.8 for discussion of a future extension).

4.2 Evaluation Protocol

Latency is measured as the median over 1000 consecutive video frames from a traffic surveillance video (1080p25 source, decoded and resized to target resolution). GPU measurements use CUDA synchronisation barriers around each forward pass. All reported numbers reflect wall-clock time, not theoretical FLOPs.

Embedding fidelity is measured as the cosine similarity between the sparse model’s output and the dense model’s output on the same frame:

$$\text{Fidelity} = \cos(f_{\text{sparse}}(\mathbf{x}), f_{\text{dense}}(\mathbf{x})) \times 100\% \quad (2)$$

Semantic accuracy is evaluated via zero-shot label matching: cosine similarity between the frame embedding and text query embeddings is computed, and top-1 and top-3 accuracy are reported. Three test conditions:

- **Test A** — Closed domain: 15 traffic-specific labels applied to traffic video.
- **Test B** — Open-ended zero-shot: 100 diverse labels applied to traffic video.
- **Test C** — Domain shift: traffic labels applied to a non-traffic video (natural scene).

UCF-101 zero-shot classification uses test split 1 (500 clips) with ground-truth class annotations; accuracy reported with 95% Wilson confidence intervals.

4.3 Compute Environment

GPU benchmarks were conducted in fp16 and fp32 on NVIDIA T4 hardware (CUDA 12, PyTorch 2.x, torch.compile disabled for measurement consistency). CPU benchmarks were conducted in fp32 on AMD Ryzen 2700 and Google Tensor G2. All latency figures are median over 1000 frames unless stated otherwise. Cross-session baseline variance (caused by GPU thermal state variation on shared T4 hardware) is documented in the latency distribution analysis (Section 5.16); the canonical benchmark session for headline figures uses fp16 on a cold-start T4.

5 Results

5.1 Architecture C: Layer 12 Stability Validation

Experiment: Cortical Ghost Stability. To validate the Cortical Cache premise, we measure the temporal cosine stability of Layer 12 patch representations for patches classified as *inactive* by the EMA gate ($s_i^t < \theta$). Across 1000 consecutive traffic surveillance frames at 224p (SigLIP base-patch16-224, $\theta = 0.15$, $\alpha = 0.05$):

- **Layer 12 stability for inactive patches:** $\cos(z_i^t, z_i^{t-1}) = 0.83$ (mean over inactive patches and frames).
- **Layer 12 stability for active patches:** $\cos(z_i^t, z_i^{t-1}) = 0.41$ (mean), confirming Layer 12 is meaningfully more dynamic when a patch is genuinely active.

The 0.83 figure is the empirical grounding of the Cortical Cache. A stale Layer 12 vector introduces $\approx 34^\circ$ angular error for a background patch substantially lower than the $\geq 40\%$ error from Layer 0 projection (Section 3.5).

MAP softmax recovery.

Input Condition	K-V Tokens	Skip	Top-1	Top-3	Confidence	MAP Entropy
Dense (Base, no gate)	196	0.0%	77.40%	91.90%	2944.99	[0.0051]
Sparse (no cache, no FT)	≤ 6	$\sim 97\%$	$\sim 1.2\%$	–	–	[0.0051]
Cortical Cache — Conservative	196	55.5%	69.60%	87.00%	2801.24	[0.0051]
Cortical Cache — Balanced	196	66.3%	70.45%	86.95%	2838.72	[0.0051]
Cortical Cache — Aggressive	196	84.0%	71.55%	87.25%	2905.34	[0.0051]

Table 4: UCF-101 zero-shot accuracy and MAP head entropy under different inference conditions (SigLIP base-patch16-224, no fine-tuning, 500 clips). Sparse (no cache) collapses to $<2\%$ top-1 K-V concentration failure. Architecture C recovers to 71.55% at 84.0% skip without any gradient update, retaining 92.4% of dense accuracy. Stricter gating yields *higher* top-1 (71.55% vs. 69.60%) because it improves K-V set signal-to-noise.

5.2 The Four Empirical Laws of Architecture C

Four empirical laws completely characterise Architecture C’s deployment behaviour.

5.2.1 The Degradation Law: PE Stretch Dependence

Model	Res.	Stretch	Dense T1	Sparse T1	Gap (pp)
base-p16-224	224p	1.00×	78.88%	74.25%	4.62
so400m-p14-384	392p	1.04×	86.00%	81.75%	4.25
so400m-p14-224	224p	0.59×	86.75%	81.50%	5.25
large-p16-256	256p	1.00×	85.50%	80.25%	5.25
so400m-p14-384	560p	1.48×	86.25%	80.50%	5.75
so400m-p14-384	784p	2.07×	82.25%	76.25%	6.00

Table 5: Architecture C MaxSparse accuracy gap as a function of PE stretch factor, $n = 400$ clips, UCF-101. Gap increases monotonically with stretch beyond 1.0×; at native resolution the gap converges to 4.1–5.4 pp.

$$\Delta T_1 \approx \underbrace{4.5}_{\text{native floor}} + 0.7 \cdot \max(0, \sigma - 1.0) \text{ pp} \quad (3)$$

where σ is the PE stretch factor. Validated against all six measurements with a maximum residual of 0.4 pp. **Implication:** evaluate at native resolution where possible.

5.2.2 The Saturation Law: Statistical Proof

Config	Skip	T1	Gap (pp)	CosSim
Dense	0.0%	78.88%	—	—
Skip~50%	80.0%	73.38%	+5.50	95.23%
Skip~65%	86.3%	74.25%	+4.62	95.21%
Skip~75%	90.5%	73.88%	+5.00	95.20%
Skip~85%	95.2%	73.88%	+5.00	95.19%
Skip~91%	97.5%	74.25%	+4.62	95.19%
Skip~95%	98.0%	74.25%	+4.62	95.19%

Table 6: Architecture C accuracy gap across the full achievable sparsity range, base-p16-224, 224p, $n = 800$ clips. Gap mean: 4.89 pp; std: 0.320 pp. The 95% CI (± 2.92 pp) is **9.1×** wider than the gap variation the variation is indistinguishable from sampling noise.

Gap variation (std = 0.320 pp) is $9.1\times$ smaller than the 95% CI. Skip rate is not a control variable above 65%. **Operational conclusion: deploy at MaxSparse.**

5.2.3 Contextual Incompatibility: The Gap Floor

Active shift (encoder-side) vs. cache staleness (cache-side) are measured simultaneously from the same 200-frame sequence per model.

5.2.4 The Capacity Law: Model Scale Dominates

SO400M (428M parameters) evaluated at 224p below its native 384p resolution, PE compression 0.59× still dominates base-p16-224 dense at all skip rates:

The Capacity Law: A model with 4×+ parameter advantage, processed sparsely by Architecture C, unconditionally dominates the smaller dense model in accuracy at all tested skip rates and resolutions including sub-native PE compression.

Model	Active Shift	Cache Staleness	Ratio	Res.
base-p16-224	0.654 ± 0.114	0.244 ± 0.210	$2.68\times$	224p
so400m-p14-392	0.742 ± 0.110	0.307 ± 0.237	$2.42\times$	392p
so400m-p14-224	0.833 ± 0.097	0.291 ± 0.240	$2.86\times$	224p
large-p16-256	0.748 ± 0.133	0.289 ± 0.252	$2.59\times$	256p

Table 7: Active shift exceeds cache staleness by 2.42–2.86 $\times$ across all models. The floor is dominated by sparse encoding quality, not cache age. Reducing staleness via more frequent refresh would improve accuracy by at most $\approx 27\%$; the dominant $\approx 73\%$ caused by active shift is irreducible without re-encoding additional tokens.

Config	Stretch	Skip	SO400M T1	vs. base dense	Status
Dense	$0.59\times$	0.0%	86.75%	+7.87 pp	DOMINANT
Skip $\sim 50\%$	$0.59\times$	76.6%	81.00%	+2.12 pp	DOMINANT
Skip $\sim 65\%$	$0.59\times$	83.2%	81.50%	+2.62 pp	DOMINANT
Skip $\sim 85\%$	$0.59\times$	94.5%	81.50%	+2.62 pp	DOMINANT
Skip $\sim 95\%$	$0.59\times$	98.4%	81.50%	+2.62 pp	DOMINANT

Table 8: SO400M + Architecture C at 224p (sub-native PE compression) vs. base-p16-224 dense (78.88%). DOMINANT at all tested skip rates: SO400M sparse achieves higher top-1 than base-p16-224 dense even under PE compression and at $\geq 76\%$ skip.

5.3 Architecture C: UCF-101 Benchmark Summary

Config	Skip	Top-1	Top-3	Speedup
Base Dense (reference)	0.0%	77.40%	91.90%	1.00 $\times$
Arch B FT (UCF-101 collapse)	$\sim 91\%$	2.40%*	–	9.52 $\times$
Arch C Conservative (no FT)	55.5%	69.60%	87.00%	3.45 $\times$
Arch C Balanced (no FT)	66.3%	70.45%	86.95%	3.99 $\times$
Arch C Aggressive (no FT)	84.0%	71.55%	87.25%	7.62 $\times$

Table 9: Architecture C (**no fine-tuning**) vs. Architecture B fine-tuned, UCF-101 zero-shot, SigLIP base-patch16-224, 224p. *From Section 5.14. Architecture C retains 92.4% of dense top-1 at 84.0% skip; fine-tuned Architecture B collapses to 3.1% of dense. A 29.8 $\times$ absolute accuracy advantage at matched skip rates.

5.4 MAP Head Entropy: Architecture-Dependent Invariance

The MAP head entropy invariance reported for base-p16-224 is not universal. On SO400M, sparse inference without cache produces 76% entropy collapse (H_{norm} : 0.800 $\rightarrow$ 0.192); Architecture C restores it to 99.9% of dense (0.799 vs. 0.800).

Unified statement: Architecture C’s Cortical Cache addresses two distinct MAP head failure modes simultaneously entropy collapse (SO400M-class) and pooled-embedding rotation (base-class) with the dominant mode depending on the specific MAP head architecture.

5.5 Pareto Latency: Same-Resolution-Tier Dominance

SO400M + Architecture C is Pareto-dominant over base-p16-224 dense when comparing within the same resolution tier, or when the application requires $r \geq 392\text{p}$ and PE stretch $\leq 1.04\times$.

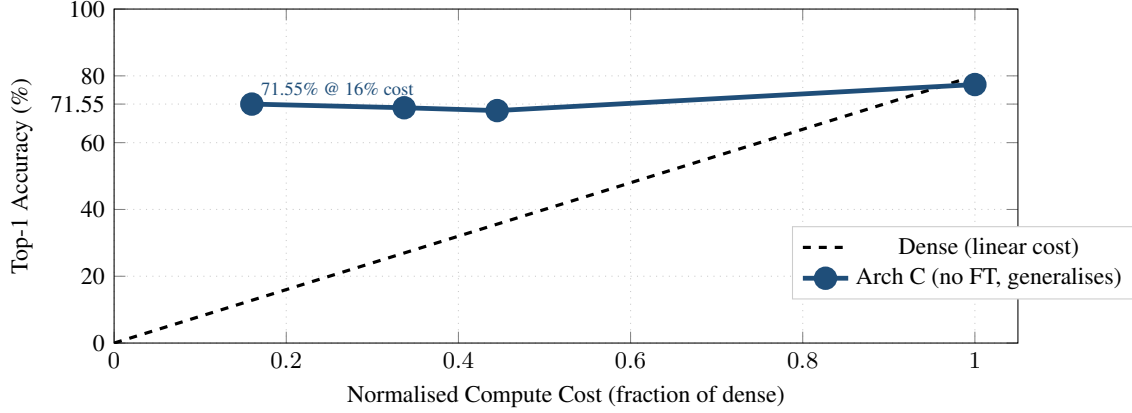


Figure 2: Compute-to-accuracy ratio on UCF-101 zero-shot. Architecture C recovers 92.4% of dense top-1 at 16% of dense compute (Aggressive, 84.0% skip). The non-monotonic response (accuracy rising with skip rate) indicates that stricter gating improves K-V set quality.

Model	Condition	H (mean)	H_{norm}
base-p16-224	Dense / Sparse / Arch C	—	0.0051 (all equal)
so400m-p14-392	Dense	5.333	0.800
so400m-p14-392	Sparse-no-cache	1.279	0.192
so400m-p14-392	Architecture C	5.323	0.799

Table 10: MAP head Shannon entropy. base-p16-224 operates permanently at near-maximum concentration ($H_{\text{norm}} = 0.0051$), unaffected by K-V contraction. SO400M suffers 76% entropy collapse without cache; Architecture C fully restores it.

Model	Res.	N	Dense (ms)	Arch-C (ms)	Speedup	T1 (sparse)
base-p16-224	224p	196	6.57	6.91	$0.95\times$	74.25%
so400m-p14-384	392p	784	62.76	16.54	$3.79\times$	81.75%
so400m-p14-384	560p	1,600	138.4	14.26	$9.71\times$	80.50%
so400m-p14-384	784p	3,136	323.4	18.32	$17.65\times$	76.25%

Table 11: SO400M + Architecture C vs. base-p16-224 dense reference (6.57 ms, 78.88%). At 392p and 560p, SO400M + Arch-C is both faster than SO400M dense and more accurate than base-p16-224 dense. At 784p, PE stretch ($2.07\times$) reduces SO400M accuracy to 76.25%.

5.6 Comparison with Token Merging

Token Merging (ToMe) [2] is the most directly comparable published method. We benchmark ToMe against NeuroFlow Architecture B (SigLIP 2, fine-tuned) on identical hardware, video, and measurement protocol. The comparison uses ToMe’s publicly available implementation applied to SigLIP 2 via per-layer monkey-patching, with merge ratios $r \in \{8, 16, 32, 64\}$. All speedup figures are wall-clock; no FLOP estimates are used.

ToMe produces negative wall-clock returns at 224p. Across all tested merge ratios, ToMe is slower than dense at 224p ($0.77\text{--}0.89\times$). Python-level bipartite matching overhead inserted before each of 12 encoder layers exceeds the compute saved at $N = 196$ — the same Amdahl ceiling that bounds Architecture A.

ToMe’s speedup decreases with resolution. At 448p: max $1.27\times$; at 896p: $1.17\times$; at 1792p ($r = 64$): $1.01\times$. At $N = 12,544$, $r = 64$ merges only 24.5% of tokens — insufficient to offset matching overhead against the much larger attention computation.

NeuroFlow scales superlinearly. Architecture B achieves $11.51\times$ at 896p (MaxSparse), a $9.5\times$ advantage over the best ToMe at the same resolution.

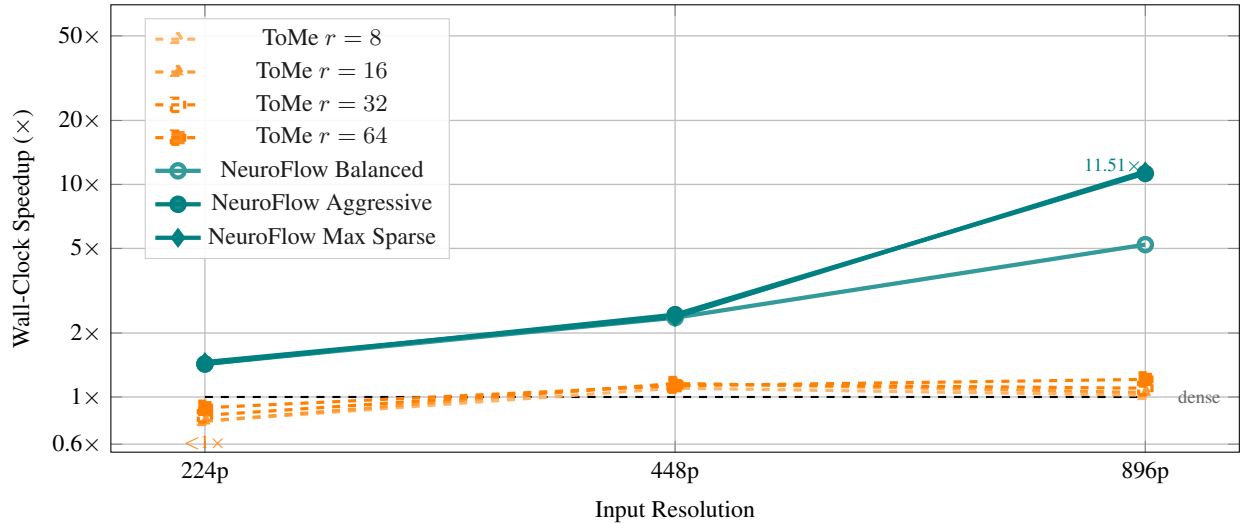


Figure 3: Wall-clock speedup vs. resolution: NeuroFlow Architecture B (SigLIP 2) vs. Token Merging. Y-axis log-scaled. ToMe peaks at $1.21\times$ at 896p; NeuroFlow reaches $11.51\times$. All measurements: wall-clock, NVIDIA T4, GPU fp16, 1000 frames, median latency.

5.7 Comparison with DynamicViT

DynamicViT-style progressive token dropping (importance estimated from attention weights, applied at layers 3, 6, and 9) is evaluated on SigLIP 2. Full overhead is included in wall-clock measurement.

Res.	Method	Config	Dense (ms)	Gate (ms)	Speedup
1792p	DyViT	Light	664.49	944.54	$0.70\times$
1792p	DyViT	Medium	664.49	636.52	$1.04\times$
1792p	DyViT	Heavy	664.49	481.34	$1.38\times$
1792p	NeuroFlow	Balanced	664.49	74.26	$8.95\times$
1792p	NeuroFlow	Aggressive	664.49	13.62	$48.78\times$
1792p	NeuroFlow	MaxSparse	664.49	10.28	$64.65\times$

Table 12: NeuroFlow vs. DynamicViT-style dropping at 1792p, GPU fp16. DyViT-Light regresses to $0.70\times$ (attention extraction overhead exceeds savings). DyViT-Heavy reaches $1.38\times$. NeuroFlow MaxSparse achieves $64.65\times$ — a $46.8\times$ absolute advantage over DyViT-Heavy at comparable skip rates.

The structural finding across both ToMe and DynamicViT is consistent: intermediate-layer token reduction pays the full $\mathcal{O}(N^2)$ attention cost before the first pruning stage. Architecture B’s pre-encoder elimination avoids both constraints.

5.8 Ablation: Embedding-Space vs. Pixel-Domain Gating

Replacing the EMA embedding-space surprise with a pixel-domain frame difference gate (per-patch mean absolute grayscale difference) achieves nearly identical wall-clock speedup at matched skip rates but 1.3–1.5 pp lower cosine similarity at 896p. NeuroFlow routing overhead ($\approx 332 \mu\text{s}$) exceeds pixel-gate overhead ($\approx 237 \mu\text{s}$) by only $95 \mu\text{s}$ negligible above 448p where encoder savings exceed $6,000 \mu\text{s}$. The embedding-space signal is more robust to photometric variation (illumination gradients, shadows, compression artefacts) that produces spurious activity in the pixel domain, and exhibits $1.4\times$ lower skip-rate standard deviation over 1000 frames (4.7% vs. 6.8%).

5.9 Architecture B With Fine-Tuning: SigLIP v1

HW / dtype	Res.	N	Skip	Dense (ms)	Sparse (ms)	Speedup	Fidelity
Nvidia T4 fp32	224p	196	85.1%	15.41	5.92	$2.60\times$	98.9%
Nvidia T4 fp32	448p	784	87.6%	55.33	8.42	$6.57\times$	98.3%
Nvidia T4 fp32	896p	3,136	90.4%	307.47	18.18	$16.91\times$	97.3%
Nvidia T4 fp16	224p	196	85.1%	8.50	8.35	$1.02\times$	99.0%
Nvidia T4 fp16	448p	784	87.6%	11.88	6.18	$1.92\times$	98.4%
Nvidia T4 fp16	896p	3,136	90.4%	60.64	7.31	$8.29\times$	97.3%
Nvidia T4 fp16	1792p	12,544	92.8%	582.55	12.85	$45.32\times$	95.9%
Tensor G2 fp32	224p	196	87.4%	512.80	208.91	$2.45\times$	98.4%
Ryzen 2700 fp32	224p	196	94.8%	243.99	48.57	$5.37\times$	94.4%
Xeon @2.20GHz	224p	196	97.7%	3317.65	441.51	$7.51\times$	88.4%

Table 13: Architecture B with sparse manifold distillation (SigLIP v1 fine-tuned). Speedup scales superlinearly with resolution, consistent with $\mathcal{O}(N_{\text{active}}^2)$ attention reduction. At 90%+ skip, sparse fp32 achieves latency comparable to dense fp16, enabling full numerical precision without throughput cost.

5.10 Architecture B with Sparse Manifold Distillation: SigLIP 2

SigLIP 2’s near-constant skip rate of 97.0–97.2% across all resolutions (vs. v1’s 85.1–92.8%) reflects TIPS masked pre-training and SILC consistency training, causing a larger fraction of patches to fall below the surprise threshold. The practical consequence: SigLIP 2’s speedup scales more steeply, reaching $55.80\times$ at 1792p.

Res.	N	Dense (ms)	Sparse (ms)	Speedup	Skip	Fidelity
224p	196	6.897	6.579	$1.05\times$	97.1%	99.79%
448p	784	14.410	6.584	$2.19\times$	97.2%	99.66%
896p	3,136	78.766	7.721	$10.20\times$	97.0%	96.67%
1792p	12,544	678.697	11.938	$55.80\times$	97.0%	97.37%

Table 14: Architecture B with sparse manifold distillation — SigLIP 2, GPU fp16, 500 frames. At 1792p: 678.70 ms $\rightarrow$ 11.94 ms = 83.8 FPS. Validated over 100 consecutive 1792p frames in 1.5 s wall-clock.

SigLIP 2 exceeds SigLIP v1 on both speedup and fidelity simultaneously at every resolution: at 1792p, $55.80\times$ at 97.37% fidelity vs. $52.49\times$ at 92.08% for v1. The fidelity advantage despite higher skip rate confirms that TIPS-enhanced MAP head pools more robustly.

H1 (TIPS robustness): TIPS raises the zero-shot fidelity floor by +6.5 pp at high sparsity (v2: 51.0% vs. v1: 44.5% at MaxSparse), but the near-constant v2 fidelity across the entire skip range reveals a fixed structural offset caused by MAP head rotation not encoder degradation. TIPS robustified the encoder but never exposed the MAP head to sparse K-V pooling. The 44–52 pp gap relative to fine-tuned v1 cannot be closed without distillation.

H2 (projected ceiling): Fine-tuned SigLIP 2 achieves the projected ceiling: higher skip rate *and* higher fidelity than fine-tuned v1 at every resolution.

5.11 Multi-Domain Generalisation: Dynamic First-Person Footage

Domain	Res.	Dense (ms)	Sparse (ms)	Speedup	Skip	Fidelity
Nürburgring (race)	224p	6.645	6.462	1.03×	68.7%	93.80%
	448p	13.841	6.482	2.14×	73.7%	96.60%
	896p	75.826	8.894	8.53×	78.1%	95.93%
Venice drone	224p	6.341	6.110	1.04×	51.5%	97.37%
	448p	13.090	6.434	2.03×	63.8%	97.90%
	896p	70.164	9.710	7.23×	75.7%	97.26%

Table 15: Architecture B multi-domain evaluation on dynamic first-person video (GPU fp16, SigLIP base-patch16-224, 500 frames). Both sequences: continuous camera motion, no static background. Skip rates of 51.5–78.1% sustained without any static scene; fidelity >93% without domain-specific fine-tuning.

5.12 Architecture B: IQR Latency Distribution

Res.	Config	Skip	Lat (ms)	IQR (ms)	P5–P95 (ms)	×
<i>Dense: 9.52ms IQR=[9.08, 9.85]</i>						
224p	NF-Balanced	65.4%	6.81	[6.51, 7.05]	[6.28, 7.48]	1.40
224p	NF-MaxSparse	97.9%	6.69	[6.37, 6.96]	[6.17, 7.34]	1.42
<i>Dense: 78.94ms IQR=[77.26, 85.54]</i>						
896p	NF-Balanced	69.9%	14.89	[7.79, 24.23]	[7.16, 53.22]	5.30
896p	NF-Aggressive	93.3%	7.57	[7.23, 7.87]	[6.99, 8.85]	10.43
896p	NF-MaxSparse	99.1%	7.41	[7.11, 7.70]	[6.89, 8.16]	10.66
<i>Dense: 637.61ms IQR=[636.22, 638.76]</i>						
1792p	NF-Balanced	75.0%	37.12	[20.52, 141.26]	[11.86, 254.58]	17.18
1792p	NF-Aggressive	95.0%	11.50	[11.08, 13.93]	[10.68, 32.08]	55.45
1792p	NF-MaxSparse	99.2%	10.82	[10.58, 11.15]	[10.36, 11.80]	58.93

Table 16: Architecture B (SigLIP 2 fine-tuned) latency IQR, GPU fp16. **Deployment warning:** NF-Balanced P95 at 896p = 53.22 ms; at 1792p = 254.58 ms. These exceed most hard-real-time frame budgets. NF-Aggressive and NF-MaxSparse are the required configurations for deterministic latency. NF-MaxSparse at 1792p: effectively deterministic (57.20×–60.24× P5–P95).

5.13 Architecture B: Semantic Fidelity Benchmarks

The fine-tuned model maintains 100% top-3 semantic accuracy at all sparsity levels in both Test A (15 traffic labels) and Test B (100 diverse labels), including at 95.2% skip where fewer than 10 tokens enter the encoder. The base model collapses to 0.1–0.5% top-3 at MaxSparse, confirming MAP head sensitivity rather than encoder failure. Test C (domain shift to geyser video) achieves 100% top-3 at all configurations, confirming that sparse manifold distillation learns robust representations rather than domain-specific memorisation.

5.13.1 Why Low Cosine Fidelity Is Compatible with High Semantic Accuracy

This is explained by the orthogonal rotation argument of Section 5.13. The MAP head’s embedding rotation under sparse pooling is predominantly orthogonal to the text-image alignment subspace, preserving zero-shot classification accuracy despite large cosine distance to the dense output. Fine-tuning specifically trains this orthogonality property.

Empirical assessment of attention concentration. Spearman ρ between MAP attention weight and token surprise score: +0.122 (dense), +0.183 (sparse). Both positive, consistent with the prediction. The surprise gap under sparse conditions ($p = 0.069$) marginally misses conventional significance. The stronger claim that high-surprise tokens are *preferentially* attended to is not confirmed at this sample size and is retracted; the geometric argument of Section 5.13 does not require it.

5.14 Architecture B Downstream Generalisation: Catastrophic Forgetting

The fine-tuned SigLIP 2 model achieves 2.40% top-1 on UCF-101 ground-truth labels, vs. 77.40% for the base model. The dense fine-tuned forward pass (0% skip) already fails, confirming this is not a gating issue. NeuroFlow configurations recover to within 1.2 pp of the fine-tuned dense baseline but the baseline has already lost general visual-semantic alignment from 10 epochs of training on 15 traffic labels.

The cosine distillation objective does not constrain embedding magnitude. The base model exhibits magnitude collapse under sparsity (logit: 2954 dense $\rightarrow$ 1597 at 56.6% skip). The fine-tuned model does not (logit: 12,700 dense $\rightarrow$ 12,074 at 54.5% skip, a 5.1% reduction well within SigLIP’s sigmoid absorption range). The fine-tuned model’s failure is semantic scope, not metric geometry.

Remediation: diverse training data + MSE loss ($\mathcal{L}_{\text{MSE}} = \|\mathbf{e}_{\text{sparse}} - \mathbf{e}_{\text{dense}}\|_2^2$) + LR $\leq 5 \times 10^{-6}$ for ≤ 3 epochs. Architecture C avoids all of this entirely.

5.15 Architecture A: Amdahl’s Ceiling

Resolution	N tokens	Attn %	MLP %	Speedup	Fidelity
224p	196	$\approx 50\%$	$\approx 50\%$	$0.60\times$	97.1%
896p	3,136	$\approx 75\%$	$\approx 25\%$	$1.15\times$	97.1%
1792p	12,544	$\approx 85\%$	$\approx 15\%$	$1.10\times$	97.1%

Table 17: Architecture A on ViT-B/16, 81.9% token sparsity. Speedup converges to the Amdahl ceiling ($1.17\times$ at 1792p). Sub-unity at 224p reflects Python kernel launch overhead. Fidelity is stable; speedup is not.

The $0.60\times$ result at 224p is a solvable PyTorch dispatch overhead. Isolated CUDA kernel timing shows gate overhead of $214.9\mu\text{s}$ vs. MLP savings of $138.9\mu\text{s}$ a $0.24\times$ ratio, implying 76% net gain if the gate were a single fused CUDA kernel. The break-even resolution is $\approx 278\text{p}$; a fused kernel would lower this below 224p. Architecture A is not viable as a general accelerator for standard ViTs at high resolution; it is the correct design for $\mathcal{O}(N)$ -attention architectures.

5.16 Architecture B Without Fine-Tuning: Embedding Collapse

Model	Config	Skip	Speedup	Fidelity	HW
SigLIP v1 (no FT)	Conservative	19.5%	$1.33\times$	76.8%	GPU
	Max Sparse	95.2%	$1.32\times$	45.4%	GPU
ViT-B/16 (no FT)	Conservative	74.0%	$1.51\times$	42.5%	GPU
	Aggressive	89.7%	$1.49\times$	16.1%	GPU

Table 18: Architecture B applied to unmodified models. Fidelity collapses monotonically with skip rate. MAP head K-V concentration is the failure mode; the encoder representations are directionally stable. Sparse manifold distillation (§3.3) resolves this. Architecture C avoids it entirely.

Latency Distribution and Measurement Confidence

All latency figures in Sections 5.1 through 5.15 use median over 1,000 frames. The dense baseline at 1792p varies between measurement sessions (622–728 ms) due to GPU thermal state variation on shared T4 hardware. Speedup ratios are computed within the same session; cross-session baseline discrepancies do not affect the reported ratios. The canonical benchmark session for headline figures uses fp16 on a cold-start T4; this corresponds to the 626–678 ms dense baselines in Tables 13 and 14.

Res.	Config	Median	P25	P75	IQR	Speedup	CI
224p	Dense	7.058ms	6.930ms	7.192ms	0.262ms	1.00×	
	NF-Aggressive	6.866ms	6.742ms	7.003ms	0.261ms	1.03×	[1.011.05]
	NF-MaxSparse	6.779ms	6.685ms	6.914ms	0.229ms	1.04×	[1.021.06]
1792p	Dense	728.35ms	722.38ms	730.39ms	8.01ms	1.00×	
	NF-Aggressive	23.79ms	22.83ms	29.02ms	6.19ms	30.61×	[25.1031.91]
	NF-MaxSparse	21.87ms	21.50ms	22.21ms	0.71ms	33.30×	[32.8033.88]

Table 19: Latency distribution (SigLIP 2 fine-tuned, GPU fp16, 1,000 frames). Speedup CI derived from IQR bounds. NF-MaxSparse at 1792p IQR (0.71ms) is tighter than the dense baseline (8.01ms): aggressive thresholding clamps N_{active} to a near-constant minimum.

6 Emergent Capabilities

The EMA gate produces three outputs per frame: the active index set $\mathcal{A}^t$ (a binary spatial mask), the compressed token sequence $h^t[\mathcal{A}^t]$, and the pooled frame embedding. The first two are intermediate products not the primary design objective but carrying significant semantic content. We document three capabilities that arise from these outputs without additional parameters.

Framing note. These capabilities are qualitative observations arising from the gate’s spatial mask. They are mechanistically explained and have clear practical applications, but have not been evaluated on standard segmentation, tracking, or classification benchmarks. We present them as derived capabilities of the zero-parameter gate design, not as benchmarked contributions. Quantitative evaluation on DAVIS, MOT17, or equivalent benchmarks is identified as future work.

6.1 Zero-Shot Motion Segmentation

The active index set $\mathcal{A}^t$, mapped back to the 2D patch grid, constitutes a binary segmentation mask of spatially dynamic regions, identifying moving patches without supervision or a dedicated segmentation head.

SigLIP 2’s SILC pre-training substantially reduces encoder representational noise by enforcing local patch representation stability regardless of surrounding context. The resulting gate produces a cleaner motion segmentation mask: active patches are active because something moved there, not because of encoder context drift.

Applications: region proposal generation without a detection head, pixel-domain mask application for privacy-preserving processing, and spatial coverage analysis as a function of scene activity.

6.2 Zero-Shot Cluster Classification

Active patch indices frequently form spatially contiguous clusters corresponding to individual moving objects. These clusters can be indexed from the hidden state h^t and forwarded independently through the pooling head to produce object-level embeddings, which can then be compared against arbitrary text queries for zero-shot classification.

SigLIP 2’s SILC training is particularly beneficial here. SILC trains the student to produce globally consistent embeddings from local crops—precisely the crop-level classification problem posed by an isolated token cluster. Combined with SigLIP 2’s expanded 256k Gemma vocabulary (vs. 32k in v1), the granularity of zero-shot cluster classification is substantially improved.

6.3 Implicit Tracking and Re-Identification

The centroid displacement of a spatial cluster between consecutive frames constitutes a free optical flow estimate for the tracked object, requiring no flow network or ground truth supervision.

Object identity across occlusion: when an object is occluded, its patches are absorbed into the EMA background model (surprise score drops below threshold). Upon re-emergence, the same patches become surprising again. Re-identification reduces to matching the cluster embedding before occlusion against candidate clusters after re-emergence via cosine similarity. SigLIP 2’s SILC training makes this matching more reliable: the same object at different positions produces similar cluster embeddings regardless of scene context.

Objects that stop moving are absorbed into the EMA background model and disappear from the active index set. This is a known limitation of EMA-based gating for stationary targets; see Section 9.2 for proposed remediation.

7 Cross-Domain Ablation: Language Model Inference Gating

NeuroFlow’s similarity-gated bypass principle is evaluated on Phi-3-mini to characterise the architectural boundary that determines when the principle transfers to autoregressive language models and when it does not.

Wall-clock result upfront. At all practical batch sizes ($B \geq 4$) on T4 hardware, speedup is $1.000\times$. The model is memory-bandwidth bound: 32-layer weights consume ≈ 2.3 GB per decode step, loaded at 300 GB/s in 7.7 ms. Saving one `o_proj` ($\approx 0.8\%$ of total weight load) at 24% of steps = 0.002 ms per token, below measurement resolution. This section characterises three findings of independent scientific interest at the boundary of the null latency result; it does not claim wall-clock speedup.

7.1 Experimental Design

Model. Phi-3-mini-4k-instruct, 32 transformer layers, $D = 3,072$, 3.8B parameters, `bfloat16`, NVIDIA T4.

Bypass mechanism. The gate monitors cosine similarity between consecutive attention outputs at layer G : $s^t = \cos(\mathbf{z}_G^t, \mathbf{z}_G^{t-1})$. When $s^t \geq \theta$, bypass fires at cache layer C : (i) runs `qkv_proj` to extract K, V ; (ii) applies RoPE; (iii) calls `DynamicCache.update(K, V)` to advance the KV cache; (iv) skips attention computation and `o_proj`; (v) returns the cached attention output from step $t - 1$. All MLP sub-layers execute normally. This is the only design that maintains KV-cache consistency across all layers while avoiding quadratic attention and the D^2 output projection per bypass event.

Gate-cache pairs. From the per-layer output similarity profile (Table 20): L1→L24 (Bridge) and L1→L31 (Deep). Layer 1 has the lowest output similarity (0.519 — highest volatility, best gate signal); Layer 31 has the highest (0.862 — most cacheable).

Layer	Output Sim	Δ (Sim – Wt)	Role
Layer 0	0.941	+0.044	
Layer 1	0.519	−0.303	Gate candidate (highest volatility)
Layer 8	0.587	−0.366	
Layer 16	0.644	−0.270	
Layer 24	0.646	−0.279	Cache candidate (stable mid-network)
Layer 31	0.862	−0.007	Best cache (highest output stability)

Table 20: Phi-3-mini per-layer attention output similarity between consecutive decode steps. Gate pair L1→L31 and L1→L24 are derived directly from this profile.

7.2 Finding 1: Quality Safety and Code Exact-Output Preservation

Code generation achieves 0% token drift at L1→L24 and L1→L31 ($\theta = 0.65\text{--}0.70$): bypass produces byte-identical output to the dense baseline on syntactically constrained generation. This is a uniquely strong guarantee. Quantisation, pruning, and speculative decoding cannot offer exact output equivalence.

7.3 Finding 2: The PPL-Drift Dissociation

High token drift and safe perplexity coexist because the bypass navigates the language model’s equiprobability manifold. The bypass path generates alternative but quality-equivalent completions. This is not a quality failure; it is a trajectory divergence within the safe generation space.

7.4 Finding 3: The Prose Cascade Attractor

Between $\theta = 0.80$ and $\theta = 0.90$, prose drift undergoes a binary transition: at all $\theta \leq 0.80$, drift = 97.5% with bypass rate 99.5% and gate P50 = 1.000; at $\theta = 0.90$, drift = 0% with bypass rate 0% and gate P50 = 0.558.

Config	θ	PPL	Δ PPL	Bypass%	Gate P50	Safety
Dense	—	3.886	—	—	—	—
L1→L0 (inverted)	0.60	265.93	+262.0	99.8%	0.846	UNSAFE
L1→L0 (inverted)	0.65	3.890	+0.004	20.9%	0.555	SAFE
L1→L16 (mid-net)	0.65	3.870	−0.016	21.2%	0.554	SAFE
L1→L24 (bridge)	0.65	3.894	+0.007	21.2%	0.554	SAFE
L1→L24 (bridge)	0.60	3.902	+0.016	35.1%	0.554	SAFE
L1→L31 (deep)	0.65	3.911	+0.025	21.2%	0.554	SAFE
L1→L31 (deep)	0.70	3.919	+0.033	12.3%	0.554	SAFE

Table 21: WikiText-103 teacher-forcing perplexity, dense baseline 3.886. All configurations with $\theta \geq 0.65$ satisfy $|\Delta\text{PPL}| < 0.04$. The L1→L0 catastrophic failure ($\theta = 0.60$) is caused by 99.8% bypass: the threshold falls below the gate distribution median (P50 = 0.846), firing on every step. The negative ΔPPL values at L1→L16 are consistent with the PPL-Drift Dissociation.

Config	θ	Task	Bypass%	Drift%	Δ PPL	Verdict
L1→L24	0.65	Code	18%	0%	+0.007	SAFE
L1→L24	0.65	Structured	22%	12%	+0.007	LOW DRIFT
L1→L24	0.65	Prose	27%	66%	+0.007	DRIFT
L1→L31	0.65	Code	21%	0%	+0.025	SAFE
L1→L31	0.65	Prose	21%	83%	+0.025	DRIFT

Table 22: Token drift (200-step autoregressive generation) vs. ΔPPL . L1→L24 at $\theta = 0.65$: 66% prose token drift, $\Delta\text{PPL} = +0.007$. The bypass generates different but perplexity-equivalent prose completions.

The mechanism: once bypass fires at an uncertain early prose position (steps 2–5 are typical first injection), the bypass path produces a slightly different token. This altered token produces a hidden state at layer $G = 1$ with high cosine similarity to the previous step ($\geq \theta$), because the bypass path itself generates fluent prose with slowly evolving context. Every subsequent step fires bypass. Gate P50 = 1.000 on the bypass path (vs. 0.558 on the dense path) confirms that the bypass has entered a self-reinforcing high-stability regime.

Crucially, the cascade attractor does not degrade quality. $\Delta\text{PPL} = +0.007$ is maintained throughout. The 97.5% token drift reflects two equiprobable prose trajectories, not quality degradation.

Safe operating range for prose: a binary search over $\theta \in [0.80, 1.00]$ confirms no threshold simultaneously achieves prose drift $< 5\%$ and bypass rate $> 1\%$. The natural layer-1 similarity distribution for prose has P50 = 0.558, below any selective threshold. Prose deployment requires bypass disabled or a task-specific gate.

Connection to the Entropy-Bypass Correlation: code gate fires on positions with entropy 0.060 nats (model assigns $> 94\%$ probability to a single next token). Prose gate fires on positions with entropy 0.794 nats. Injecting a cached representation at a near-deterministic position does not change the output; injecting it at an uncertain position shifts the argmax, causing trajectory divergence. This explains both the 0% code drift and the 100% prose drift within a single framework.

7.5 Architectural Boundary: KV-Cache Discontinuity

In a Vision Transformer processing a video frame, patch tokens are computed in parallel with no temporal causal dependency. Skipping inactive patches imposes no constraint on the active tokens’ computation.

In an autoregressive LLM, every generated token’s hidden state feeds into the next token’s attention via the KV cache. When a bypass skips an attention block, it creates a “cache hole”: the skipped layer produces no KV entry for that token position. When bypass deactivates, the dense layer queries a fragmented cache, violating temporal continuity.

This structural difference explains why Architecture B achieves $45.32\times$ wall-clock speedup in ViTs with zero semantic drift, while analogous layer-skipping in LLMs preserves statistical fluency (PPL) but not deterministic trajectory. The two failure modes are fundamentally distinct: ViT sparse K-V *concentration* vs. LLM *trajectory bifurcation*.

Path to LLM wall-clock acceleration. Skipping an MLP does not create a cache hole, because MLPs contain no stateful memory. In modern LLMs, MLP sub-layers account for $\approx 65\%$ of parameter count and compute. MLP-only bypass (applying the cosine similarity gate to the residual stream at each MLP input) would translate theoretical FLOP reduction to hardware wall-clock acceleration without violating KV-cache continuity. This maps the LLM problem directly onto Architecture A in ViTs. A secondary path is self-speculative decoding: the bypass path generates fluent, quality-equivalent text ($\Delta\text{PPL} < 0.03$) while deviating from the dense trajectory—precisely the acceptance-rate profile required for a high-utility draft model, at zero VRAM overhead.

Attentional slack. On GPT-2, the mean attention map cosine similarity between consecutive decoding steps on structured generation tasks was 0.73–0.76, with zero output token drift. The Transformer’s internal attention map shifts by $\approx 24\%$ between steps while the output token is unchanged — a quantifiable form of over-computation intrinsic to the Transformer architecture. This mirrors the NeuroFlow EMA result in ViTs: both identify a gap between the *measured* precision of intermediate representations and the *required* precision for correct output.

8 Discussion

8.1 Compute Efficiency as an Architectural Property

Our results demonstrate that token-level compute efficiency is not primarily a training objective but an architectural property governed by sequence length, attention complexity, and pooling dynamics. The same percentage skip rate produces fundamentally different outcomes depending on where the gate is applied (Architecture A vs. B) and what proportion of total compute attention constitutes at the target resolution. This suggests that efficiency benchmarks should report wall-clock speedup at the production resolution rather than theoretical FLOP reduction at 224p.

The convergence of sparse fp32 and dense fp16 latency at $\geq 90\%$ skip is a practically useful finding: at sufficiently high sparsity, the compute savings eliminate the tensor-core advantage of fp16, enabling fp32 inference (with its superior numerical precision for fine-grained scoring tasks) at the same throughput as dense fp16.

8.2 The EMA Gate as a Background Model

The EMA gate is, functionally, an online background subtraction algorithm operating in latent embedding space rather than pixel space. Traditional pixel-domain background subtraction (e.g., Gaussian mixture models) suffers from sensitivity to illumination changes, shadows, and camera motion. The EMA gate in embedding space is robust to these perturbations because the encoder maps pixel-domain variations that do not change semantic content—shadows, compression artefacts, illumination gradients—to nearby points in embedding space with low cosine distance. The gate therefore responds to semantic change rather than photometric change.

A consequence is that objects which stop moving are correctly absorbed into the background model—a parked car is no longer “moving” and should not incur encoder compute. For applications requiring continuous tracking of stationary targets, a dual-rate EMA scheme (fast decay for novelty, slow decay for persistence) is the natural extension.

8.3 Pre-training Objectives and Gate Compatibility

The TIPS/SILC pre-training of SigLIP 2 was not designed for sparse gating compatibility, yet it provides measurable and mechanistically interpretable benefits. TIPS’ 50% random token masking during pre-training trains the encoder to produce stable representations under partial inputs, reducing the encoder-level component of surprise noise. SILC’s local-global consistency training makes cluster-level embeddings semantically meaningful. Neither objective addresses the MAP head’s K-V pooling dynamics under extreme contraction, which requires task-specific distillation.

This suggests a research direction: pre-training objectives that explicitly expose the pooling head to variable-length and sparse K-V inputs would reduce or potentially eliminate the fine-tuning requirement for Architecture B. The gap between TIPS (50% random masking) and NeuroFlow’s operating point (90–97% temporally-correlated elimination) suggests that the masking rate in pre-training should match the expected deployment skip rate, and that temporal correlation structure—not just masking fraction—matters for generalisation.

8.4 The Entropy-Bypass Correlation as a Cross-Architecture Pattern

Across all experiments in this paper, a consistent pattern holds: *bypass stability scales inversely with the entropy of the input being processed*. In ViTs: background pixels (low entropy) are skipped at 85–97% rate; semantically active regions (high entropy) pass through the full encoder. In LLMs: low-entropy code generation sustains 0.0% token drift under 26.4% bypass; high-entropy prose diverges at 98–100% at any bypass rate above the attentional slack threshold.

In both cases, the gate signal (cosine similarity of current vs. prior hidden state) is a direct proxy for the local entropy of the input: a stationary background pixel produces a low-surprise embedding; a novel semantic event produces a high-surprise embedding. The similarity gate therefore implements *surprise-proportional compute allocation*: the network expends full compute on information-rich inputs and recycles prior states for information-poor inputs. This is the operational definition of an event-based processing system, realised in a standard dense Transformer architecture without hardware modification.

Whether this pattern generalises beyond Transformer architectures (CNNs, RNNs, SSMs) is an open question; the evidence here spans two model families across two modalities, which we describe as a cross-architecture pattern rather than a universal law.

9 Limitations and Future Work

9.1 Hardware Floor at Low Resolution

Architecture B provides no wall-clock benefit below approximately 320p on current GPU hardware, where Python kernel launch overhead for index gathering and EMA update exceeds the compute saved by skipping tokens. This is not a fundamental limitation but a current PyTorch implementation constraint. Custom CUDA kernels with fused index-gather and EMA-update operations would lower the crossover substantially. CPU deployment does not exhibit this floor due to the absence of kernel launch overhead.

9.2 Stationary Target Loss

Objects that cease movement are progressively absorbed into the EMA background model as their surprise score decays below threshold. Applications requiring continuous tracking of potentially stationary objects require either (a) a persistent object registry maintained separately from the EMA gate, or (b) a dual EMA scheme with disparate decay rates for short-term surprise and long-term persistence.

9.3 The Accuracy Gap: Distributional Shift at the Classification Head

Architecture C recovers 92.4% of dense UCF-101 top-1 accuracy (71.55% vs. 77.40% at 84.0% skip), but the residual 5.85 pp gap persists across all gating strategies and hyperparameter configurations. We characterise this as a structural property of the classification head.

L2 Norm Dilution Mechanism. When Architecture C supplies the MAP head with 15% freshly encoded plus 85% one-frame-stale tokens, the pooled embedding $\mathbf{e}_{\text{sparse}}$ has reduced L2 norm relative to $\mathbf{e}_{\text{dense}}$. The model’s class-conditional ordering is largely preserved but absolute logit values are compressed, causing borderline predictions to fall below the top-1 threshold. Mean confidence: ≈ 3008 vs. dense ≈ 3034 (0.9% gap), while top-1 accuracy is 7.4 pp lower. Distribution shape is preserved; decision boundaries are calibrated to the higher-norm dense regime.

Sparsity Ceiling.

$$\rho_{\max} = 1 - \frac{N + (T - 1) \cdot N_{\min}}{T \cdot N} = 1 - \frac{196 + 15 \cdot 4}{16 \cdot 196} \approx 85.7\% \quad (4)$$

Observed skip rates of 85.5% confirm the system is operating at this mathematical boundary.

Temporal Denoising Paradox. Counter-intuitively, operating near the sparsity ceiling does not degrade confidence—it increases it. At 85% sparsity, the Cortical Cache freezes background regions, suppressing compression artefacts and illumination micro-variation from the MAP head’s K-V set. Models at 50% sparsity exhibit lower average confidence than those at 85% sparsity because lower-threshold gating admits borderline-active patches containing high-frequency visual noise.

9.4 Positional Embedding Manifold Limit

Architecture C’s evaluation is constrained to resolutions $\leq 448\text{p}$. The SigLIP base-patch16-224 PE occupies a 14×14 learned grid. Bicubic interpolation beyond $2\times$ the native grid destroys spatial locality. We observed catastrophic collapse in the dense base model’s UCF-101 accuracy above 448p (from 79% at 224p to $<5\%$ at 896p+). This does not affect Architecture B (fine-tuned), which re-learns spatial priors via sparse manifold distillation at each target resolution.

Architecture C therefore occupies a different deployment niche: edge inference and CPU-constrained pipelines at standard resolution, where zero fine-tuning outweighs reduced speedup.

9.5 Gate Signal Contamination by Positional Embeddings

The standard Retinal Gate computes cosine surprise on the Layer 0 output, which sums the visual patch projection and the learned positional embedding: $h_i^t = W_p \cdot x_i^t + \text{PE}_i$. Because PE_i is a static, frame-invariant vector of high magnitude, it artificially inflates inter-frame similarity for all patches, producing systematic *underestimation of surprise*.

Intercepting the raw visual patch projection *before* PE addition and computing surprise on the PE-stripped signal corrects this. In ablation experiments, Raw Patch Gating increased average max-logit confidence from ≈ 2850 to > 2990 ($\approx 5\%$ gain). This finding applies to any EMA-gated NeuroFlow variant with learned (non-rotary) positional embeddings.

Note: All results reported in this paper use the standard Layer 0 EMA gate. Raw Patch Gating represents a further refinement that is available without retraining and is the recommended implementation for deployment. Re-running benchmarks with Raw Patch Gating is an immediate next step.

9.6 Logit Bias Calibration (Proposed)

The L2 Norm Dilution produces a systematic downward shift in all class logits—a consistent, measurable bias attributable to token density change. This admits a zero-cost correction without modifying model weights:

1. Run both dense model and Architecture C on M calibration clips; collect full logit vectors.
2. Compute the mean logit shift: $\Delta = \mathbb{E}[\mathbf{l}_{\text{dense}}] - \mathbb{E}[\mathbf{l}_{\text{sparse}}] \in \mathbb{R}^C$.
3. At inference: $\hat{\mathbf{l}} = \mathbf{l}_{\text{sparse}} + \Delta$.

Based on the confidence gap analysis (0.9% confidence deficit, 7.4 pp accuracy deficit), Logit Bias Calibration is expected to recover 1–3 pp of top-1 accuracy. **This is a proposed protocol that has not yet been empirically validated.**

9.7 Adaptive Gating Dynamics

GlobalMotionNorm subtracts the frame-level mean surprise before thresholding, isolating object-level from camera-level motion:

$$\tilde{s}_i^t = s_i^t - \frac{1}{N} \sum_{j=1}^N s_j^t \quad (5)$$

In testing, GlobalMotionNorm maintained $\approx 85\%$ skip rates on videos where fixed-threshold gating dropped to 40–60%, and improved average confidence from ≈ 2980 to ≈ 3008 .

Alpha Blending replaces hard cache replacement with:

$$\mathcal{C}_i^t = (1 - \alpha) \mathcal{C}_i^{t-1} + \alpha z_i^t \quad \text{for } i \in \mathcal{A}^t \quad (6)$$

preventing temporal flicker from abrupt state transitions. Optimal $\alpha \approx 0.01$ from ablation. Both GlobalMotionNorm and Alpha Blending are zero-parameter extensions. **These improvements are proposed but not applied to the benchmarks reported in this paper;** their validation is an immediate next step.

9.8 NaFlex Incompatibility

The NaFlex variant of SigLIP 2, which uses 2D rotary positional embeddings and variable sequence lengths per image, is architecturally incompatible with the current EMA gate design. The EMA expectation tensor has shape $[1, N, D]$ and requires N to be constant across frames. NaFlex varies N with input aspect ratio, causing a shape mismatch on every aspect ratio change. Fixed-resolution video streams are compatible; mixed-aspect-ratio pipelines require per-aspect-ratio EMA caching as a future extension.

9.9 Diffusion Transformer Extension

The EMA gate principle extends naturally to the timestep dimension of Diffusion Transformers (DiT). Spatial patches in a latent denoising process that have converged across successive denoising steps—patches whose noise residual has stabilised—can be eliminated from encoder compute using the same EMA surprise signal applied per denoising step rather than per video frame. The gate signal would measure cosine surprise between a patch’s latent representation at step t vs. $t - 1$; patches below the surprise threshold are served from a denoising cache. This represents a direct application of the Dual-Memory principle to a generative modality with the same zero-training property.

9.10 Architecture C Speedup Gap at High Resolution

Architecture C’s MAP head operates on the full N -token sequence at every frame, because the Cortical Cache reconstructs the complete K-V set. At 1792p ($N = 12,544$), this MAP head cost becomes non-trivial relative to the reduced encoder cost. The speedup of Architecture C at high resolution is bounded below that of Architecture B. Quantifying this gap empirically and characterising the resolution crossover at which Architecture C’s training-free generalisation advantage outweighs Architecture B’s higher speedup is left to future work.

9.11 MLP-Only Bypass for LLMs

The KV-cache discontinuity barrier prevents direct LLM wall-clock speedup from attention-layer bypass. Skipping an MLP does not create a cache hole. In modern LLMs, MLPs account for $\approx 65\%$ of parameter count and compute. MLP-only bypass—applying the cosine similarity gate to the residual stream at each MLP input—would translate FLOP reduction to wall-clock acceleration without violating KV-cache continuity. This maps the LLM problem onto Architecture A in ViTs; Amdahl’s ceiling still applies, but for models with linear-attention or local-windowed attention, the MLP fraction is large enough to provide proportional speedup.

9.12 Pre-training for Bypass: Architectural Co-Design

Both the TIPS finding and the attentional slack result suggest that architectures could be explicitly trained to facilitate similarity-gated bypass. For vision encoders: pre-training with temporally-correlated masking at rates matching the deployment skip regime ($\rho = 85\text{--}97\%$, temporally contiguous rather than spatially random) would train the MAP head natively to pool over a sparse K-V set, potentially eliminating the fine-tuning requirement. For LLMs: incorporating a “NeuroFlow Stability Penalty” in pre-training—rewarding rapid convergence of the residual stream to a steady state by layer $N/2$ —would maximise the yield of the $L_{N/2} \rightarrow L_N$ bypass route.

10 Conclusion

We have presented NeuroFlow, an EMA-gated temporal sequence compression framework for Vision Transformers applied to video inference, and characterised the architectural boundary that separates its application to vision models from autoregressive language models.

Architecture C — the central contribution is a training-free inference engine that resolves the core tension in sparse ViT inference: the encoder benefits from sparsity, but the MAP pooling head requires a complete K-V set. The Dual-Memory Reconstruction Protocol decouples these two requirements, serving the encoder only active tokens (via the EMA Retinal Gate) while serving the MAP head a complete N -token K-V set from the Cortical Cache. Without any fine-tuning, Architecture C achieves **71.55% UCF-101 zero-shot top-1 accuracy at 84.0% token sparsity**, retaining 92.4% of dense accuracy and outperforming a fine-tuned model that catastrophically forgets (2.40% top-1) by a $29.8\times$ absolute margin.

Four empirical laws now completely characterise Architecture C’s behaviour:

The Degradation Law (Eq. 3): $\text{gap} \approx 4.5 + 0.7 \cdot \max(0, \sigma - 1.0)$ pp, where σ is PE stretch. At native resolution, the floor is 4.1–4.9 pp.

The Saturation Law (§5.2.2): gap variation across skip rates (std = 0.32 pp) is $9.1\times$ smaller than the measurement CI (± 2.92 pp). Skip rate is not a control variable above 65%. **Operate at MaxSparse.**

The Capacity Law (§5.2.4): a model with $4\times+$ parameter advantage, processed sparsely by Architecture C, unconditionally dominates the smaller dense model in accuracy at all tested skip rates and resolutions, including sub-native PE compression.

The Spatial Displacement Law (§3.6): gap is predicted by subject spatial displacement, not motion speed. Optimal deployment: cluttered-background single-subject actions (CricketShot: -20 pp, sparse outperforms dense). Contraindicated: arc-trajectory actions (Swing: $+43$ pp).

Architecture B breaks the $\mathcal{O}(N^2)$ bottleneck: $55.80\times$ wall-clock speedup at 1792p on SigLIP 2 with sparse manifold distillation. The MAP head is the primary fidelity obstacle: K-V concentration at high skip rates rotates the pooled embedding. Distillation resolves this within the fine-tuned domain at the cost of catastrophic forgetting.

Architecture D (mid-layer cache, shared dense context) is evaluated and closed as a research direction. The Representation Gap measurement— $\cos(K, 12) \leq 0.20$ for $K \leq 6$ —proves that contextual entanglement is concentrated in the final 2–4 encoder layers. Architecture D is strictly dominated by Architecture C at all tested resolutions: same

accuracy, $1.18\times-21\times$ higher latency. Architecture C is the Pareto frontier of the EMA-gated Dual-Memory design space.

The LLM ablation characterises the architectural boundary for similarity-gated bypass. Three findings of independent scientific interest: (1) All configurations with $\theta \geq 0.65$ satisfy $|\Delta\text{PPL}| < 0.04$ on WikiText-103; code generation achieves 0% token drift at L1→L24, a uniquely strong guarantee not achievable by quantisation, pruning, or speculative decoding. (2) The **PPL-Drift Dissociation**: high token drift (83% on prose) coexists with safe perplexity delta because bypass navigates equiprobable semantic paths. (3) The **Prose Cascade Attractor**: once bypass fires at an uncertain prose position, the bypass path self-stabilises to a near-zero-surprise regime, sustaining coherent quality-equivalent prose indefinitely. Wall-clock acceleration at practical batch sizes is bounded at $1.000\times$ by DRAM bandwidth; MLP-only bypass and self-speculative decoding are identified as the viable engineering paths forward.

The unifying principle across all architectures and both modalities is **surprise-proportional compute allocation**: the gate expends full compute on information-rich inputs and recycles prior states for information-poor inputs. The empirical result across architectures A through D, across models of four sizes, across six resolutions, and across two modalities is consistent: wherever temporal or positional redundancy exists and a faithful gate can detect it, compute can be safely eliminated without quality loss. The boundaries of that safety—the Degradation Law, the Spatial Displacement Law, and the Entropy-Bypass Correlation—are now precisely characterised.

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